

Arkajit Mukherjee

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SKILLS

Programming Languages: Python, C++, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: FastAPI, React, Tailwind CSS, Express.js
Backend & Databases: Node.js, REST APIs, MySQL, SQLite, PostgreSQL
Developer Tools: Git, GitHub, Cursor, Claude Code, OpenAI Codex

EXPERIENCE

Software Developer

Waterloo, ON

UW Orbital (University of Waterloo Satellite Design Team) 

April 2026 – Present

- Developed a full-stack command management system for the UW Orbital Ground Station team using **React** and **FastAPI**, implementing **RESTful CRUD** endpoints and real-time UI updates.
- Implemented and tested FastAPI backend endpoints for command creation and deletion with **request validation** and middleware logging for performance monitoring.
- Engineered a dynamic **React** interface that generated command input fields from backend data and synchronised frontend state with API responses.

Software Developer

Mississauga, ON

Excelitas Technologies Inc. 

Jan 2026 – May 2026

- Developed and deployed a **Node.js/Express.js** web application providing direct **CRUD** access to internal databases, replacing reliance on tools such as MySQL Workbench for non-technical staff.
- Designed **RESTful APIs** (GET, POST, PUT, DELETE) interfacing with **10+ MySQL and SQLite databases** spanning **hundreds of tables**, enabling seamless data operations across the organization.
- Improved workflow efficiency by enabling non-technical users to interact with relational data through a clean UI, reducing manual query time and errors.
- Implemented backend logic for database connections, query execution, and error handling, ensuring stable and maintainable server-side performance.

PROJECTS

Gitanalyze | *Python, FastAPI, React, PostgreSQL*

- Built a backend analysis API using **Python** and **FastAPI** to retrieve data from GitHub repositories through GitHub REST API and compute a custom health score based on repository metrics (stars, forks, issues, PRs, watchers).
- Implemented **RESTful** API endpoints with request validation, error handling, and a modular backend to analyze repository health and return structured JSON responses.
- Integrated **PostgreSQL** and **SQLAlchemy** to store repository data for trend tracking and future analytics features.

Delirium Detection Handheld | *C++, ESP32, Gemini API*

- Developed an **ESP32-based** system to automate **delirium screening**, improving assessment consistency and accessibility for ICU patients.
- Programmed multi-modal input/output interfaces using **C++** and integrated **Gemini API** to generate standardised cognitive assessment questions and log patient responses.
- Implemented structured response logging and patient data handling enabling post-assessment analysis and future server-side storage and API integration

Personal Portfolio Website | *JavaScript, HTML, CSS, Tailwind*

- Developed and deployed a responsive personal website using **HTML**, **JavaScript**, and **Tailwind CSS** to showcase projects and experiences.
- Tested across **3 device breakpoints** (desktop, tablet, mobile) ensuring consistent UX. Leveraged **Tailwind utility classes** for efficient styling.

CERTIFICATIONS

- Introduction to Front-End Development  | Meta (Coursera)
- The Arduino Platform and C Programming  | University of California, Irvine (Coursera)

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science (BASc), Computer Engineering

Sep 2025 - Apr 2030 (expected)